Lecture 09
Deep Learning

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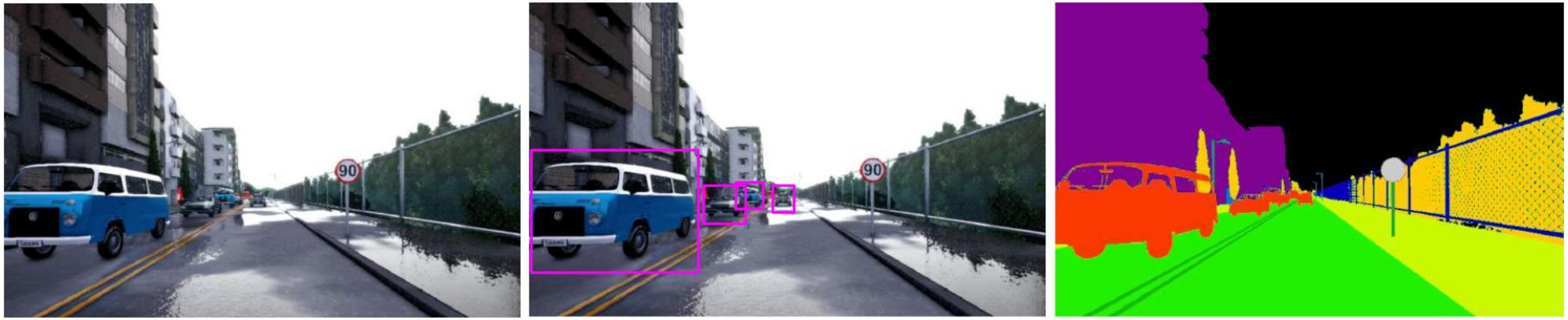
Agenda

- **Unet**
- **Speckle**
- **Noise Metrecs**
- **Ejemplo**

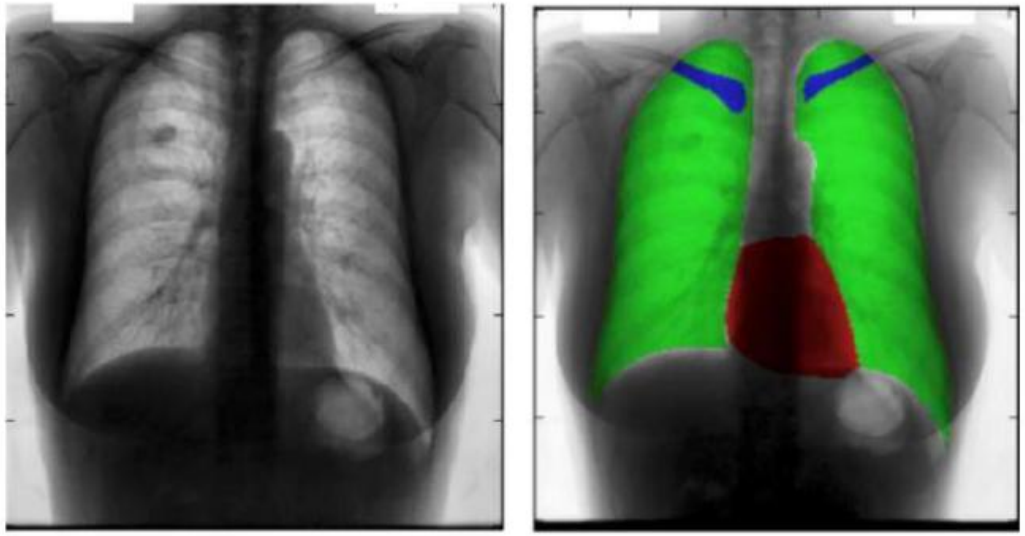
Object detection vs Semantic Segmentation



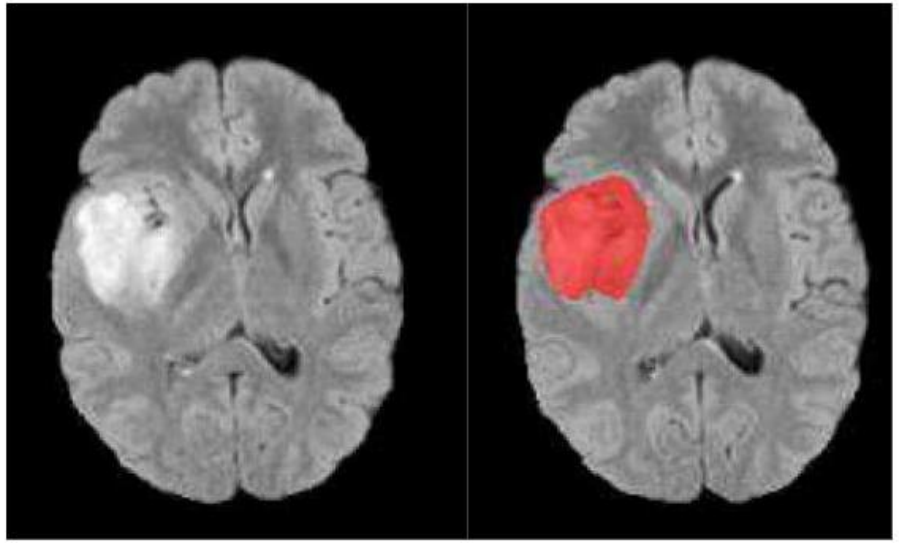
Input image | Object Detection | Semantic Segmentation

Unet

Object detection vs Semantic Segmentation



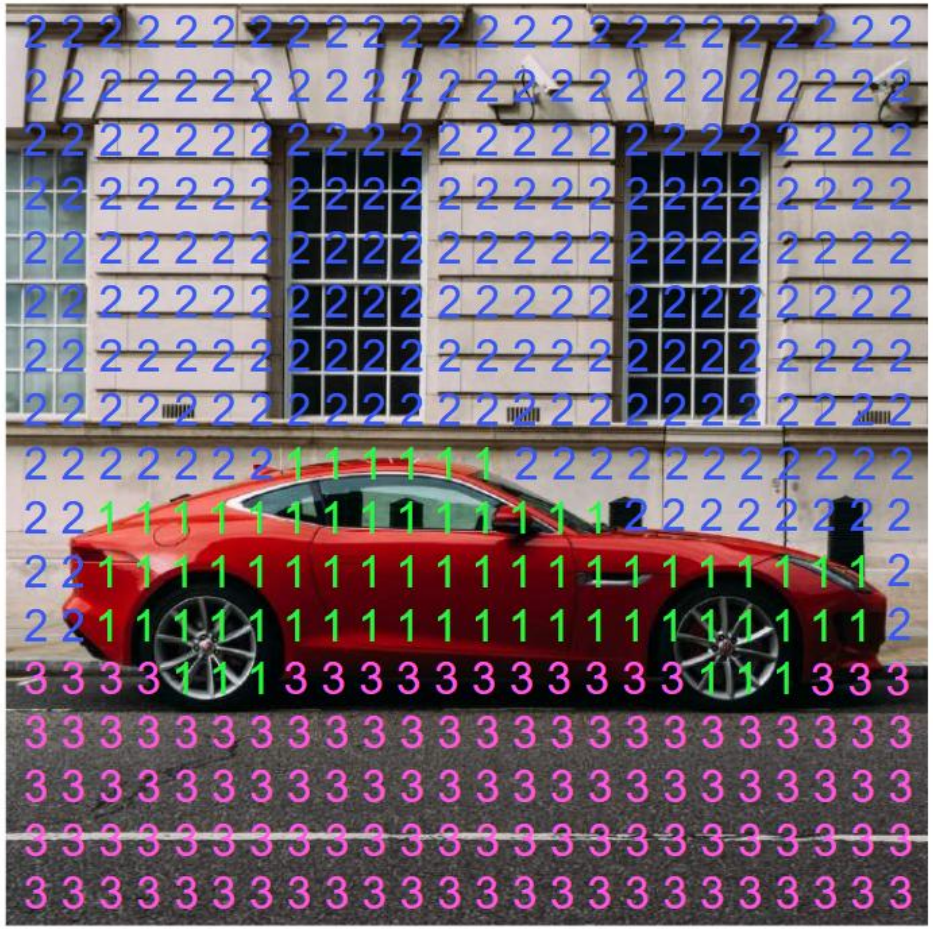
Chest X-Ray



Brain MRI

[Novikov et al., 2017, Fully Convolutional Architectures for Multi-Class Segmentation in Chest Radiographs]
[Dong et al., 2017, Automatic Brain Tumor Detection and Segmentation Using U-Net Based Fully Convolutional Networks]

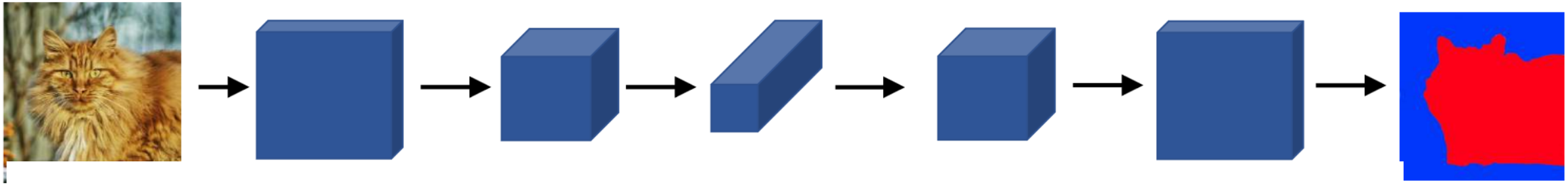
Unet



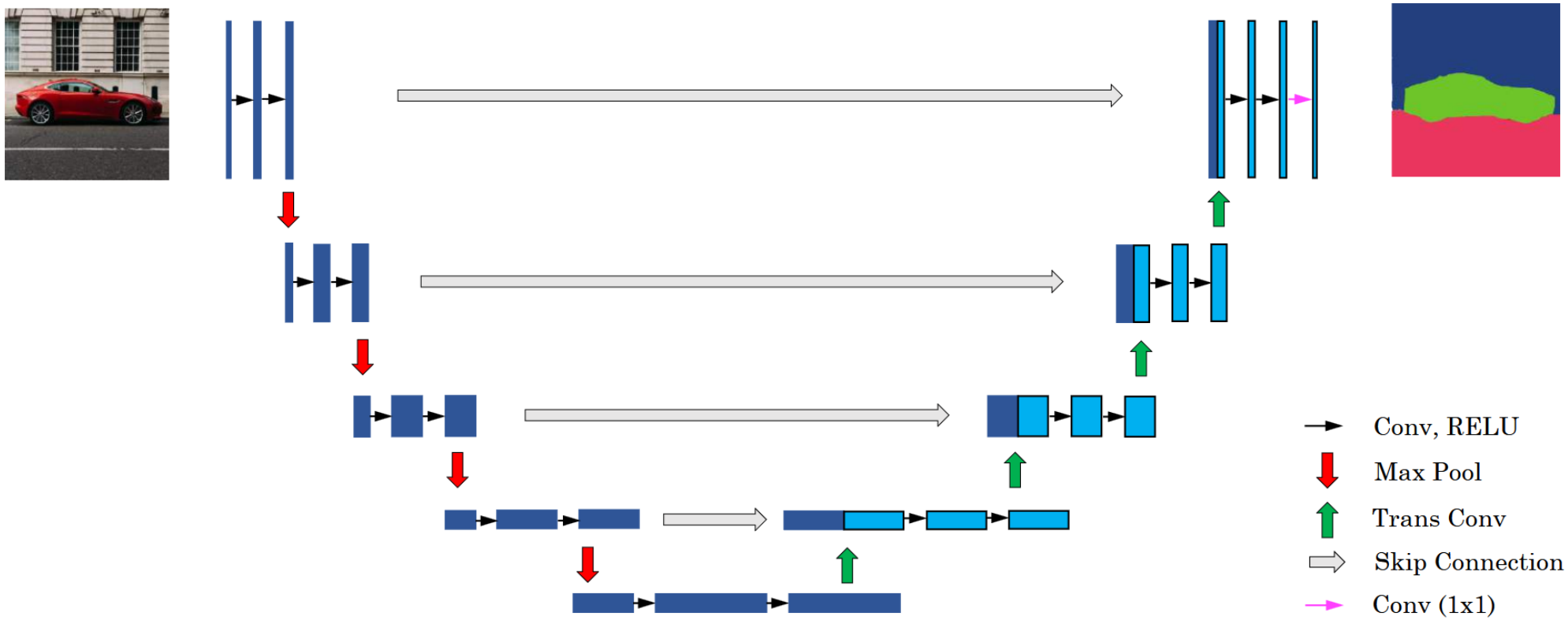
1. Car
2. Building
3. Road

Segmentation Map

Unet



U-Net



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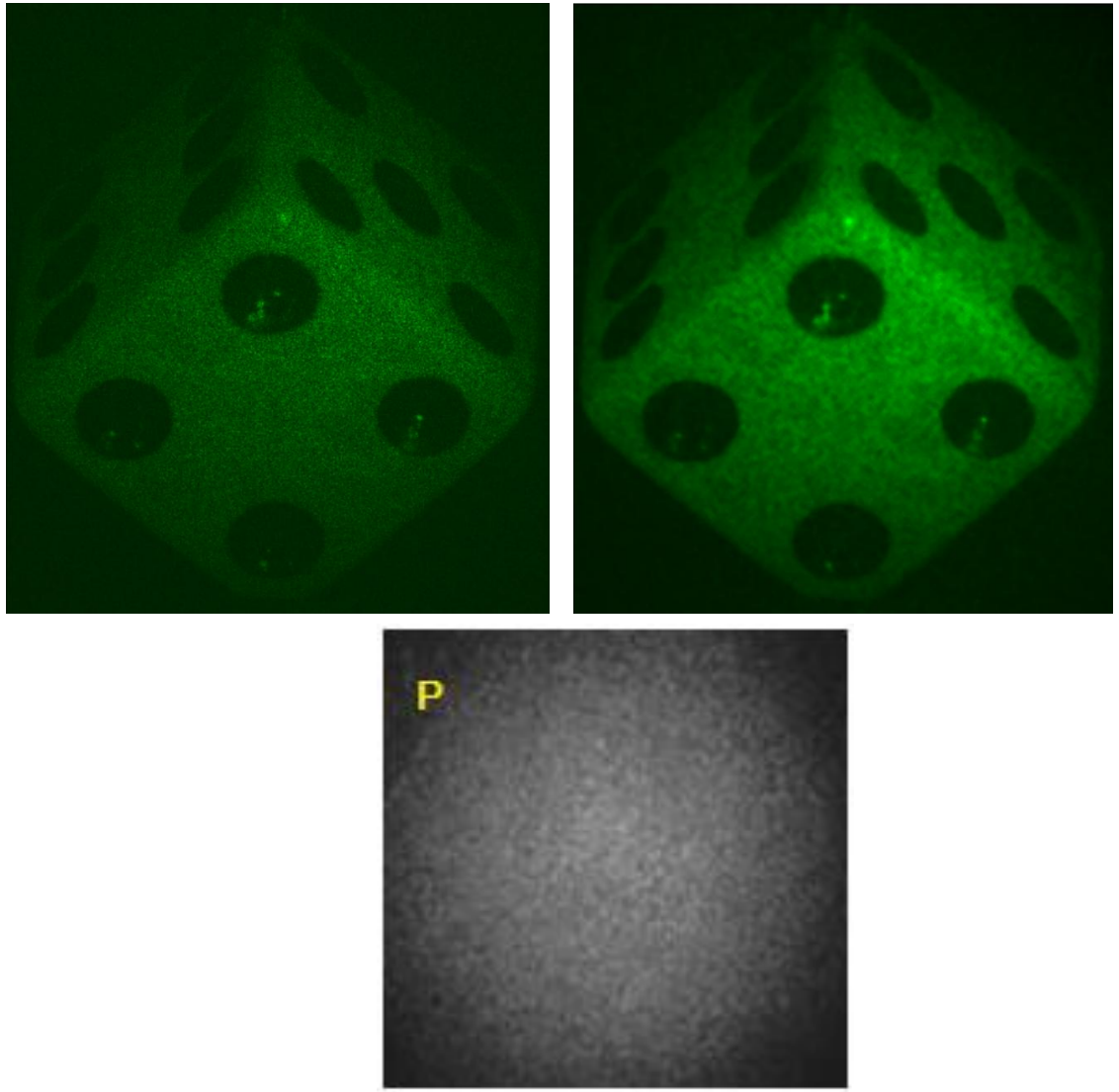
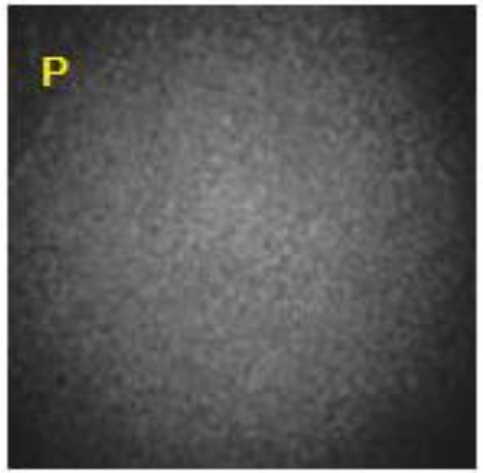
Speckle noise in coherent Images



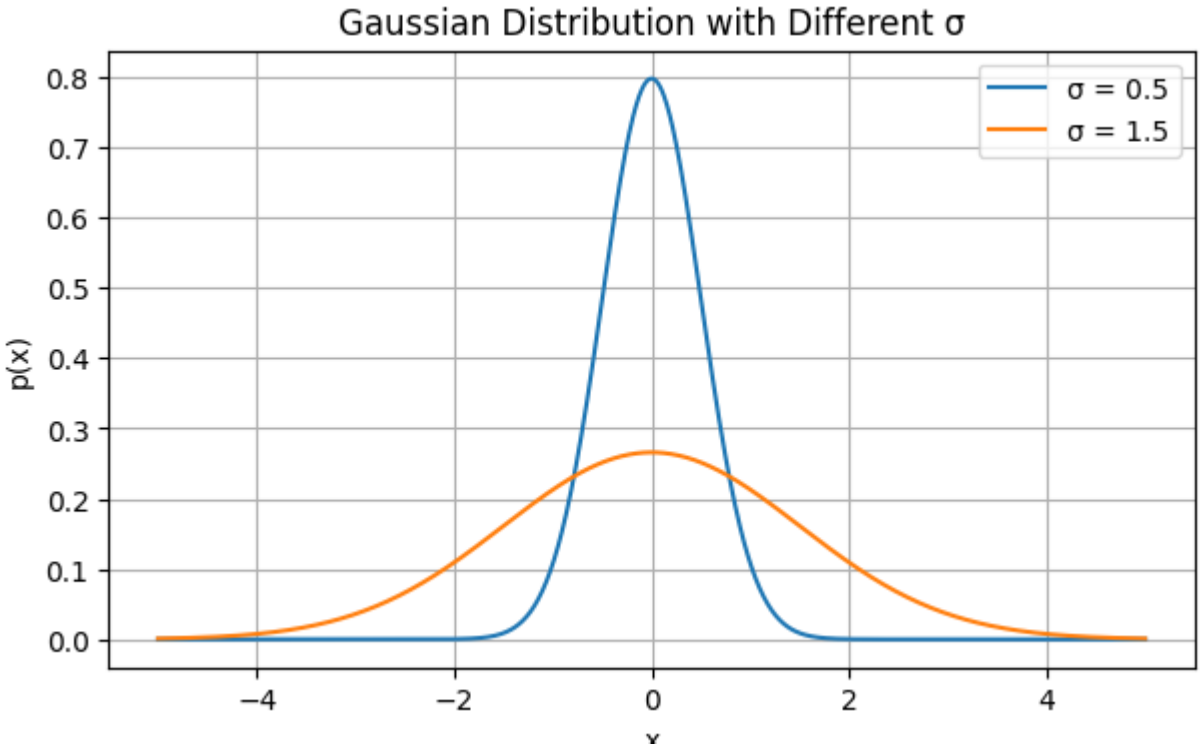
- Speckle is an inherit noise that affects any imaging techniques that utilize a coherent light as a source of illumination.
- The speckle is present as a set of dark and brightness points (granular pattern).
- The speckle contrast C is the conventional metric for measuring the speckle noise in image processing.
- C is calculated as $c = \sigma / \bar{I}$, where σ and $\bar{I}$ are the standard deviation and mean intensity, respectively, of a specific region on the image.
- The speckle noise disturbs the image.

R. Balamurugan, Int. J. Comput. Sci. Eng. Technol. 7, 163–166 (2016).

Speckle noise in coherent Images



$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} exp\left(-\frac{(x - \mu)^2}{\sigma^2}\right)$$



How can we quantify the noise?

Speckle noise in coherent Images

How can we quantify the noise?

Peak Signal-to-Noise Ratio (PSNR)

Peak Signal-to-Noise Ratio (PSNR) is a measurement used to evaluate the quality of reconstructed or enhanced images by comparing the maximum possible signal strength to the noise that affects its fidelity

It is expressed in decibels (dB) because it is a logarithmic relationship between signal and error

$$MSE = \frac{1}{MN}\sum_{x=1}^{M}\sum_{y=1}^{N}(I_D(x,y) - I_S(x,y))^2$$

Image dimensions

Pixel intensity in the clean image

Pixel intensity in the noisy image

$$PSNR = 10log_{10}\left(\frac{MAX^2}{MSE}\right)$$

PSNR	Interpretation
< 20 dB	severely degraded image
20–30 dB	visible degradation
30–40 dB	good quality
> 40 dB	almost indistinguishable from the original

Speckle noise in coherent Images

How can we quantify the noise?

Speckle Index (SI)

A speckle index, often referred to as a Speckle Contrast measures the intensity of granular noise (speckle) in imaging systems such as Synthetic Aperture Radar (SAR) or ultrasound.

$$\boxed{SI = \frac{\sigma}{\mu}}$$

σ ← Standard deviation intensity

μ ← Average intensity

$$\sigma = \sqrt{\frac{1}{MN}\sum_{x=1}^{M}\sum_{y=1}^{N}(I\,(\mathrm{x},y)\;-\mu)^2}$$

$$\mu = \frac{1}{MN}\sum_{x=1}^{M}\sum_{y=1}^{N} I\,(\mathrm{x},y)$$

Speckle noise in coherent Images

How can we quantify the noise?

Equivalent Number of Looks (ENL)

The Equivalent Number of Looks (ENL) is a critical parameter in Synthetic Aperture Radar (SAR) imaging that measures the level of speckle noise reduction, with higher values indicating lower noise and better quality. It represents the actual degree of averaging performed during imaging, often calculated in homogeneous, non-textured regions

$$\boxed{ENL = \left(\frac{\sigma}{\mu}\right)^2}$$

Standard deviation intensity

Average intensity

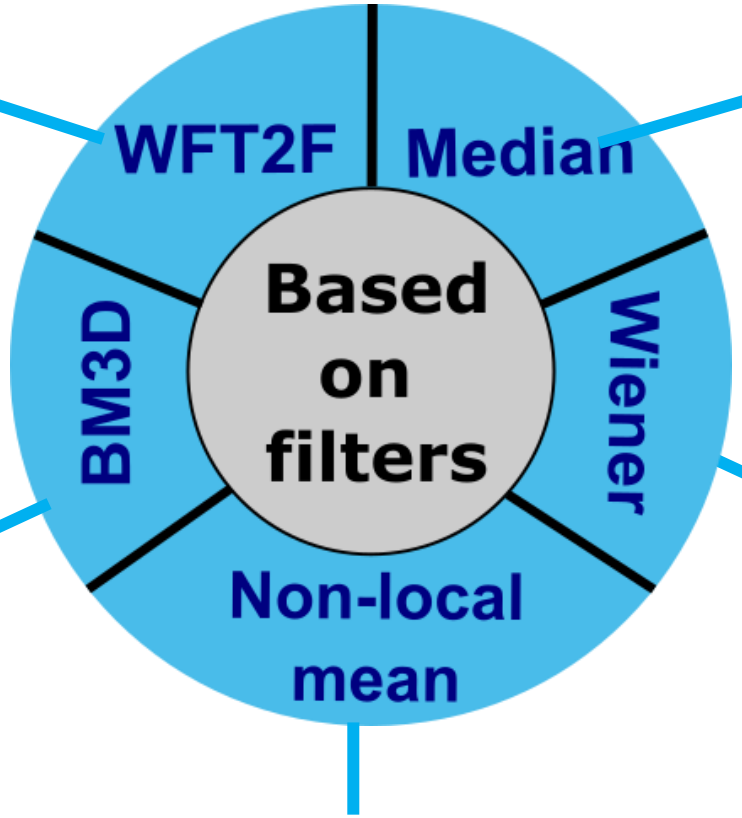
$$\sigma = \sqrt{\frac{1}{MN}\sum_{x=1}^{M}\sum_{y=1}^{N}(I\,(\mathrm{x},y)\ - \mu)^2}$$

$$\mu = \frac{1}{MN}\sum_{x=1}^{M}\sum_{y=1}^{N} I\,(\mathrm{x},y)$$

Speckle noise in coherent Images

- Frequency domain
- Local Fourier transform
- Non-stationary characteristics of the noise

Compromise image quality



- Replacing the value of pixel with an average of the pixels of similar pixels
- Scan a regions in the whole image looking for all the pixels that resemble the pixel value.

- Divide the image in group
- The groups are blocks with reference similarities
- Block are group in 3D cylinder.

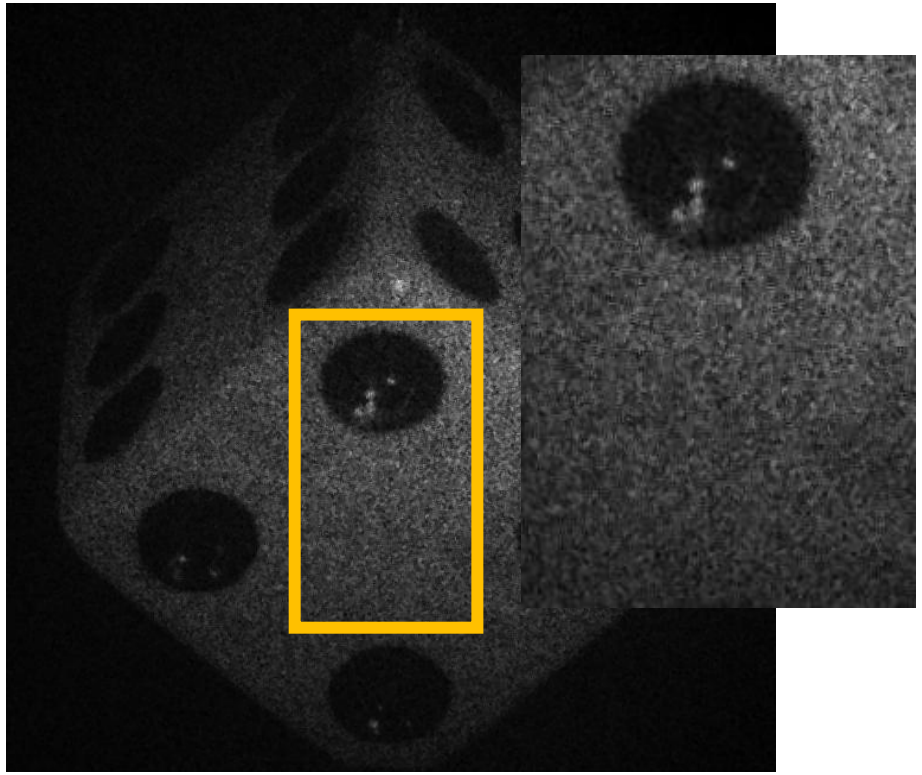
WFT2F = 2D windowed Fourier Transform Filter
BM3D = Block-matching and 3D filtering

- Median, replacing the center of kernel pixel value for the median of neighboring entries.
- Non-lineal filters

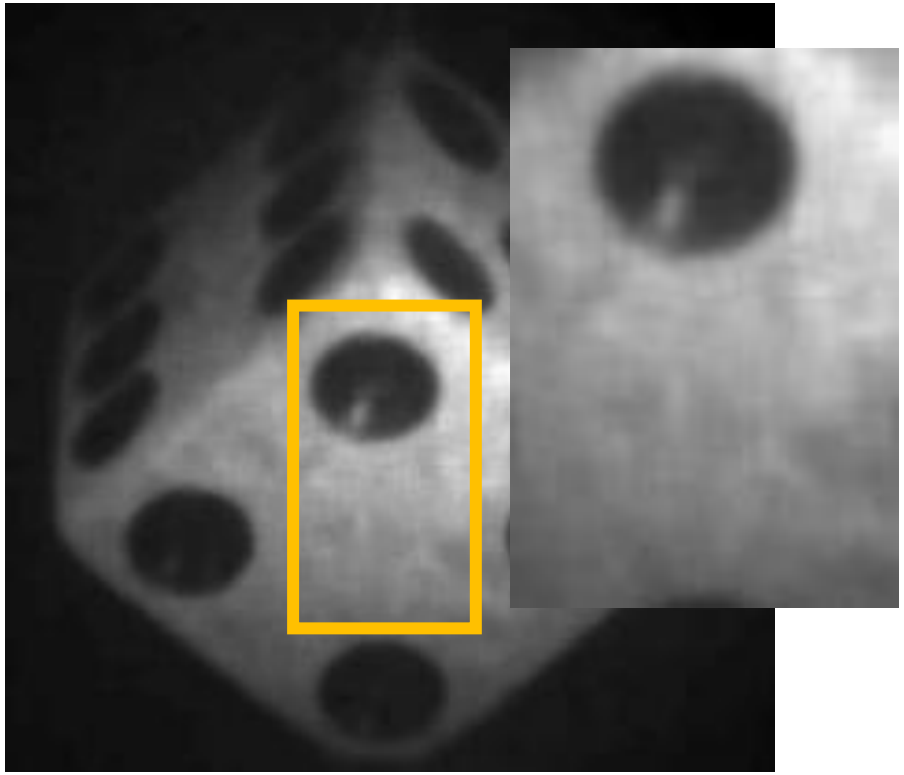
- Frequency domain
- Partial knowledge of the degradation is available
- Implement means square error minimization

Speckle noise in coherent Images

Original reconstructed amplitude DH image

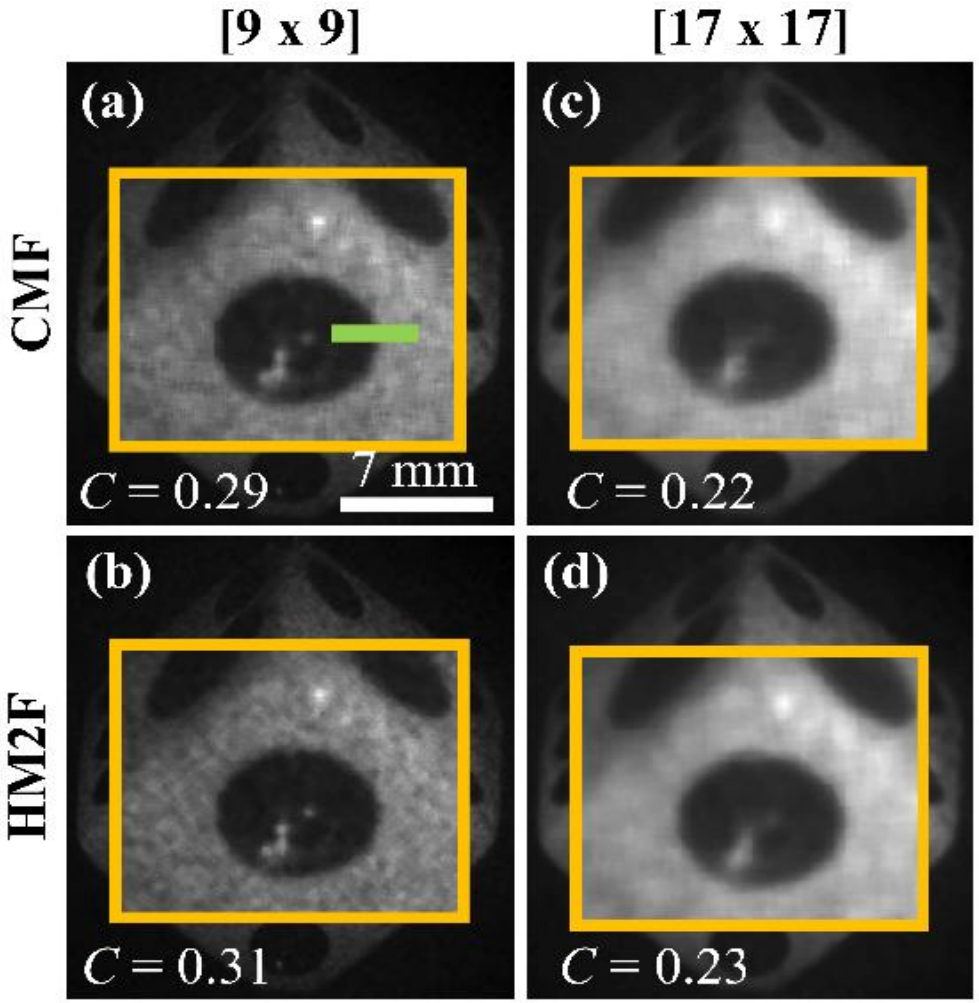


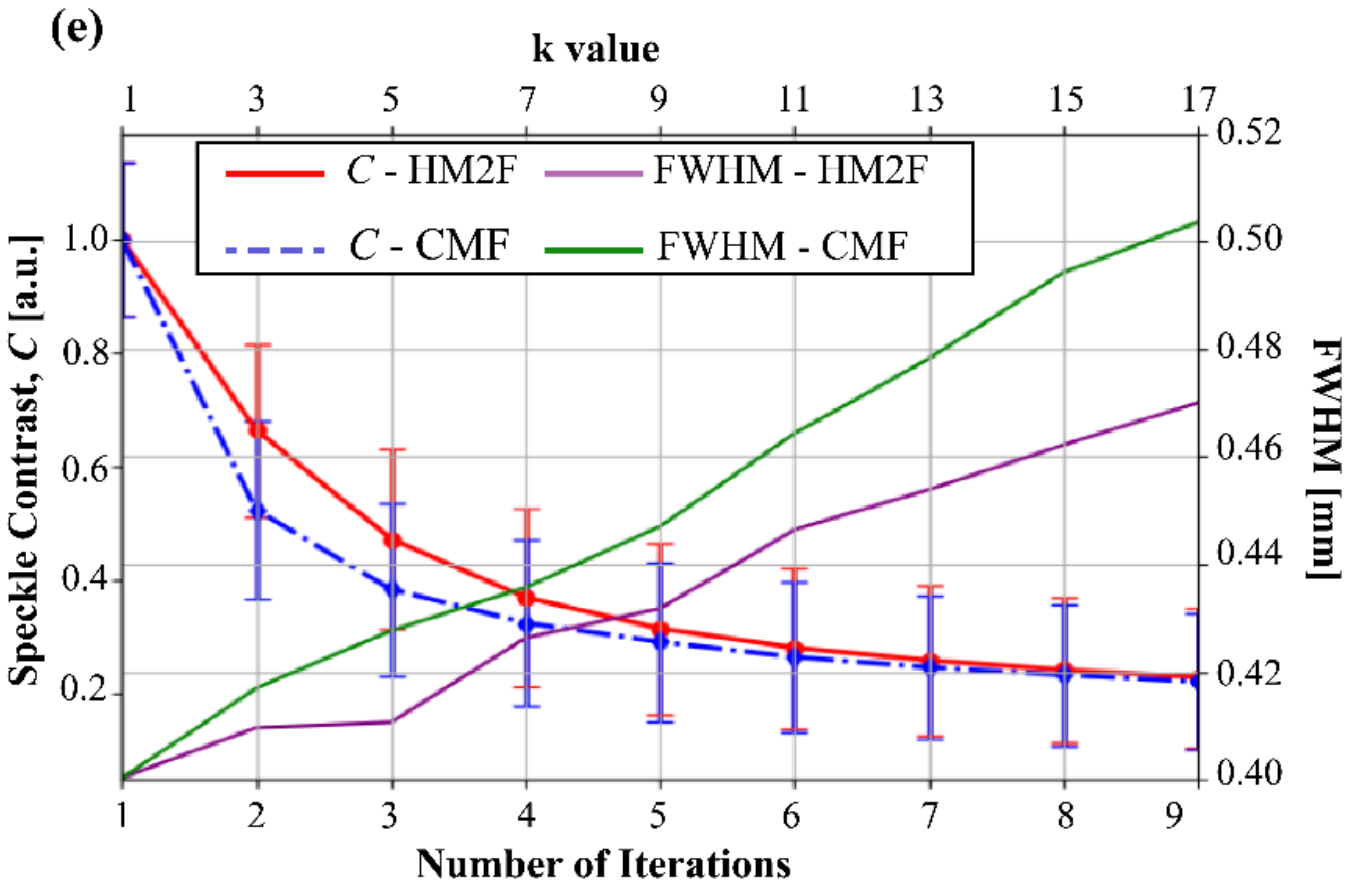
Median-filtered reconstructed DH image



The application of the median filter reduces the speckle noise as well as the resultant image is more blurred than the input imagen (i.e., reduction of the resolution limit).

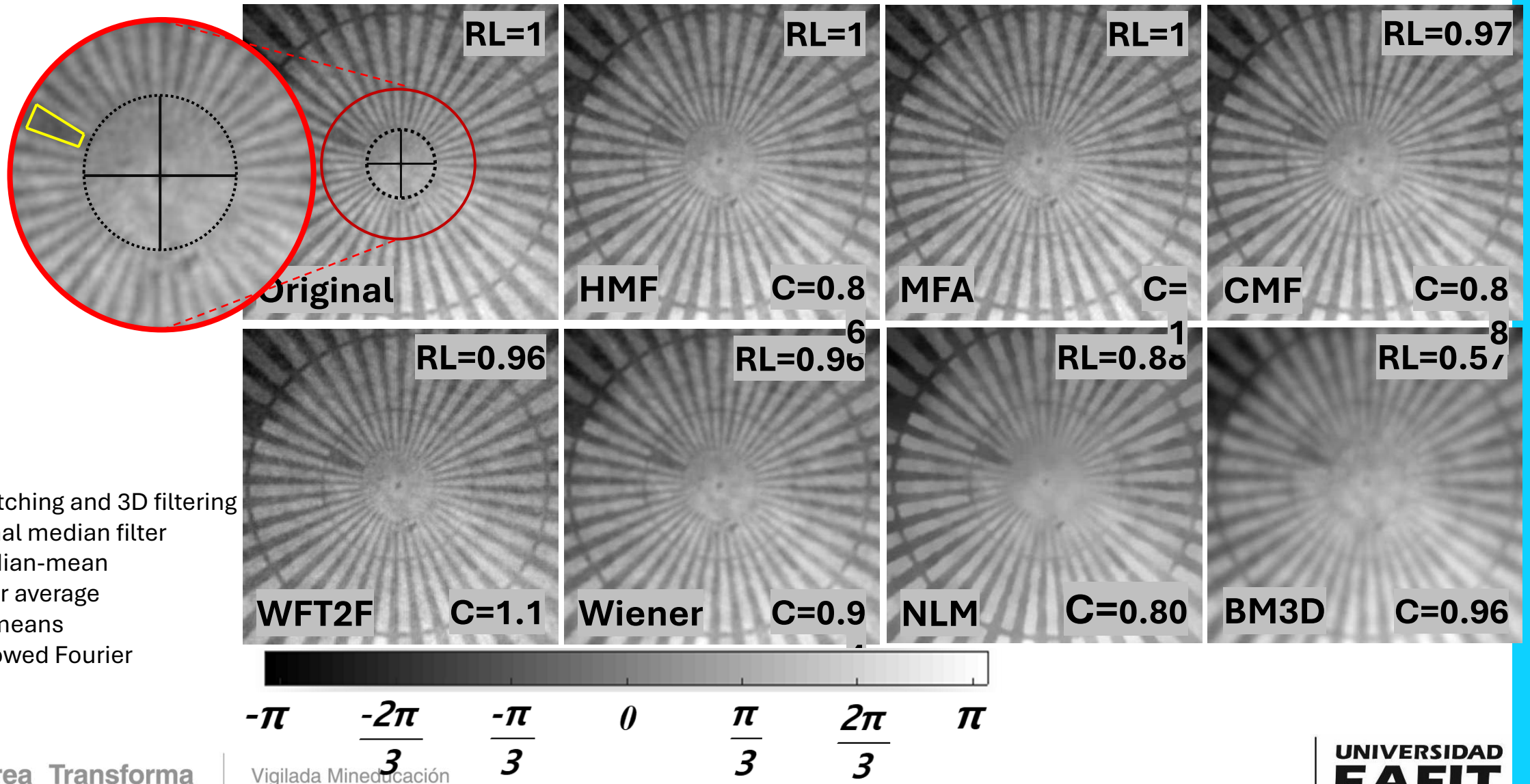
Speckle noise in coherent Images





R. Castaneda, J. Garcia-Sucerquia & Doblas, et al., Opt. Engineering 60(12), 123107 (2021).

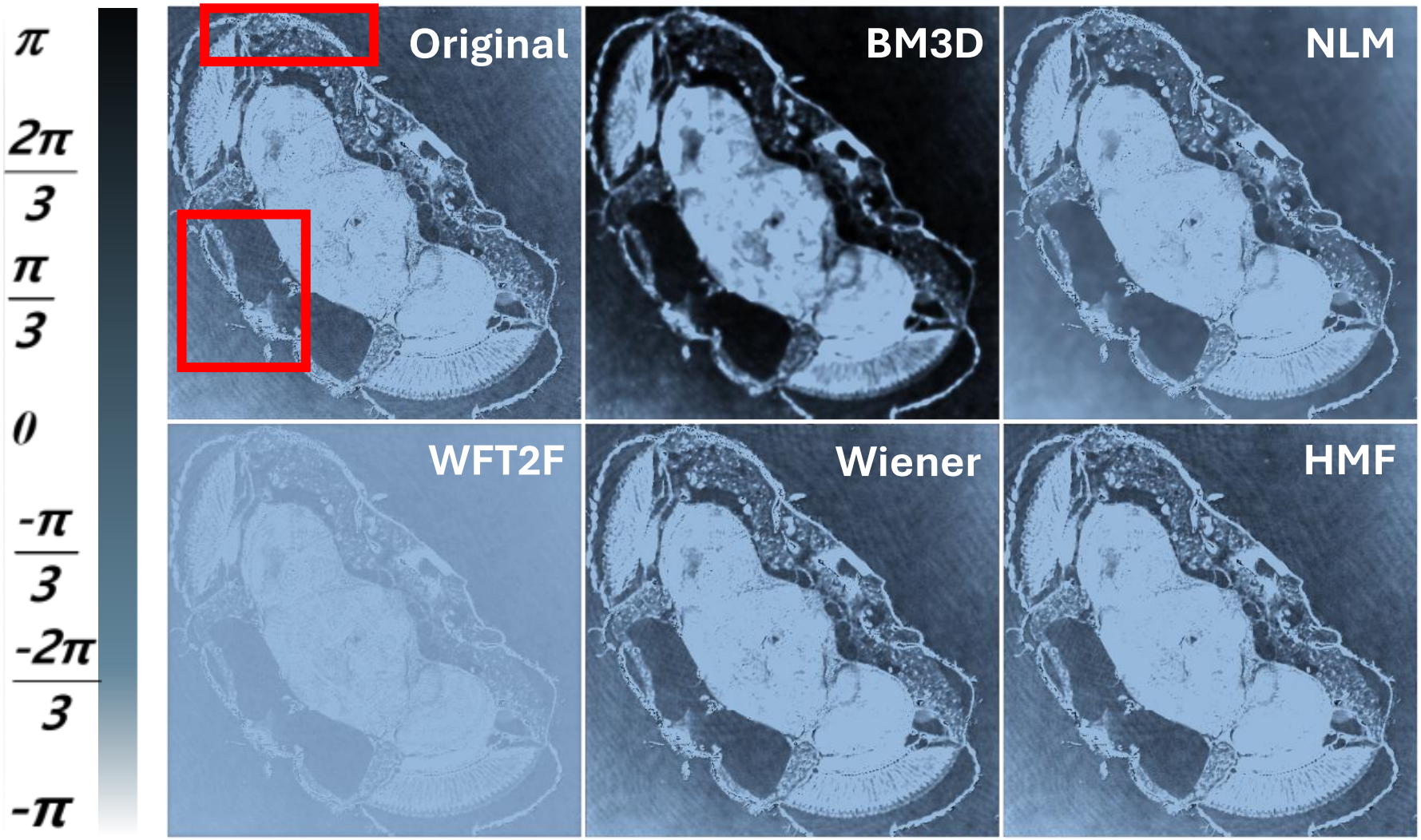
Speckle noise in coherent Images

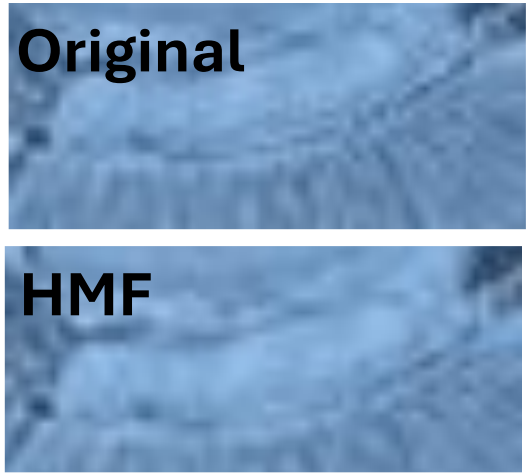


BM3D = Block-matching and 3D filtering
CMF = Conventional median filter
HMF = Hybrid median-mean
MFA = Median filter average
NLM = Non-local means
WFT2F = 2D windowed Fourier Transform

R. Castaneda, J. Garcia-Sucerquia & Doblas, et al., Opt. Engineering 60(12), 123107 (2021).

Speckle noise in coherent Images





Phase jumps	
Original	201
NLM	195
Wiener	176
Hybrid	79

BM3D = Block-matching and 3D filtering
HMF = Hybrid median-mean
NLM = Non-local means filter
WFT2F = 2D windowed Fourier Transform Filter

R. Castaneda, J. Garcia-Sucerquia & Doblas, et al., Opt. Engineering 60(12), 123107 (2021).

Unet - Example

```python
class ConvBlock(nn.Module):
    """Convolutionla block: Conv → BatchNorm → ReLU (×2)"""

    def __init__(self, in_channels, out_channels):
        super().__init__()
        self.block = nn.Sequential(
            nn.Conv2d(in_channels, out_channels, kernel_size=3, padding=1),
            nn.BatchNorm2d(out_channels),
            nn.ReLU(inplace=True),
            nn.Conv2d(out_channels, out_channels, kernel_size=3, padding=1),
            nn.BatchNorm2d(out_channels),
            nn.ReLU(inplace=True),
        )

    def forward(self, x):
        return self.block(x)
```

New class (layer-like behavior)

Arguments

Group several layers in sequence using nn.Sequential.

Convolution applies learnable filters to the image or feature map to detect local patterns

This method defines how information flows forward.

Unet - Example

```python
class SpeckleDenoiseUnet(nn.Module):
    """
```

```python
def __init__(self, in_channels=1, features=[32, 64, 128, 256]):
    super().__init__()

    # — Encoder ————————————————————————————————————————
    self.enc1 = ConvBlock(in_channels, features[0])   # 128 → 128
    self.enc2 = ConvBlock(features[0], features[1])   # 64  → 64
    self.enc3 = ConvBlock(features[1], features[2])   # 32  → 32
    self.enc4 = ConvBlock(features[2], features[3])   # 16  → 16

    self.pool = nn.MaxPool2d(2, 2)

    # — Bottleneck ————————————————————————————————————
    self.bottleneck = ConvBlock(features[3], features[3] * 2)  # 8×8
```

Feature maps

Encoder

Max Pooling

It's the most compact representation of the image. The network attempts to preserve the important information from the original image

Unet - Example

```
# — Decoder ————————————————————————————————————————————————
self.up4 = nn.ConvTranspose2d(features[3]*2, features[3], kernel_size=2, stride=2)
self.dec4 = ConvBlock(features[3]*2, features[3])   # +skip

self.up3 = nn.ConvTranspose2d(features[3], features[2], kernel_size=2, stride=2)
self.dec3 = ConvBlock(features[2]*2, features[2])   # +skip

self.up2 = nn.ConvTranspose2d(features[2], features[1], kernel_size=2, stride=2)
self.dec2 = ConvBlock(features[1]*2, features[1])   # +skip

self.up1 = nn.ConvTranspose2d(features[1], features[0], kernel_size=2, stride=2)
self.dec1 = ConvBlock(features[0]*2, features[0])   # +skip
```

Decoder

Increase the spatial resolution of the image

This is what doubles the resolution.

$$H_{out} = (H_{in} - 1) \times stride + kernel \; - \; 2 \times padding$$

The `ConvTranspose2d` layer performs learned upsampling, increasing the spatial resolution of the feature maps from 8×8 to 16×16. The result is then concatenated with the corresponding encoder feature maps (skip connection), doubling the number of channels. A convolutional block then processes this combined information to refine the reconstruction.

Unet - Example

```
# — Decoder ————————————————————————————————————————————————
self.up4 = nn.ConvTranspose2d(features[3]*2, features[3], kernel_size=2, stride=2)
self.dec4 = ConvBlock(features[3]*2, features[3])   # +skip

self.up3 = nn.ConvTranspose2d(features[3], features[2], kernel_size=2, stride=2)
self.dec3 = ConvBlock(features[2]*2, features[2])   # +skip

self.up2 = nn.ConvTranspose2d(features[2], features[1], kernel_size=2, stride=2)
self.dec2 = ConvBlock(features[1]*2, features[1])   # +skip

self.up1 = nn.ConvTranspose2d(features[1], features[0], kernel_size=2, stride=2)
self.dec1 = ConvBlock(features[0]*2, features[0])   # +skip
```

Decoder

Increase the spatial resolution of the image

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Unet - Example

Loss function a real problem

In an image denoising problem, especially with speckle noise, it is not enough for the network to "numerically approximate" the clean image.

It is also important that it preserves the visual structure of the image: edges, contours, global textures, and local relationships between pixels.

$$\mathcal{L}_{\mathrm{t}} = \alpha \mathcal{L}_{\mathrm{MSE}} + \beta \mathcal{L}_{\mathrm{SSIM}}$$

MSE: forces pixel values to be similar.

SSIM: forces the visual structure of the image to be similar.

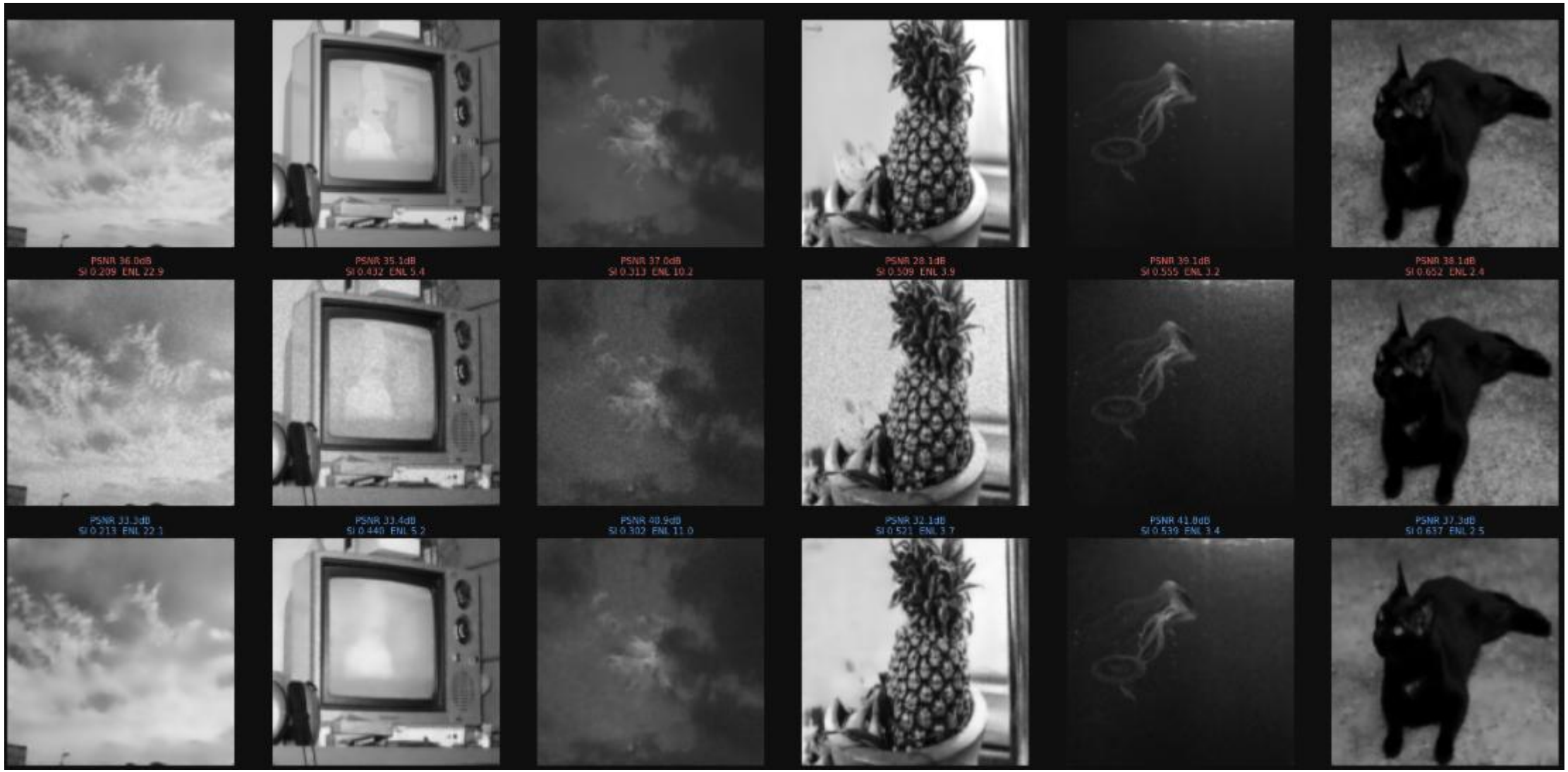
Unet - Example

$$\mathcal{L}_{\mathrm{t}} = \alpha \mathcal{L}_{\mathrm{MSE}} + \beta \mathcal{L}_{\mathrm{SSIM}}$$

MSE treats an image as if it were just a collection of numbers, pixel by pixel.

It doesn't properly understand that an image has: edges, shapes, textures, local patterns, spatial relationships between neighbors.

SSIM Structural Similarity Index Measure. It's a metric designed to measure the similarity between two images in a way that more closely approximates human visual perception. Instead of just looking at pixel-by-pixel differences, it compares: luminance, contrast, structure.

$$SSIM(\mathrm{x},\mathrm{y}) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

Unet - Example